FACULTY OF ARTS

KU LEUVEN

Computational Linguistics

Lecture 1: Introduction

Tim Van de Cruys



Tuesday, 22 September 2026

Modeling language with computers

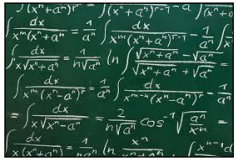
- From a **scientific** point of view
  - computational linguistics
  - study of production, comprehension, and acquisition of linguistic phenomena from a computational perspective
- From an **engineering** point of view
  - natural language processing
  - use computers to do useful things with language, e.g. question answering, sentiment analysis, machine translation

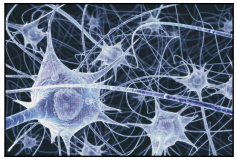


Three paradigms

- **symbolic** approach
  - knowledge-based approach to language modeling
  - handcrafted dictionaries and grammars
- **statistical** approach
  - statistics and machine learning
  - large annotated corpora, handcrafted features
- **neural** approach
  - neural networks
  - large (un)annotated corpora, *no feature engineering*







Applications

Sentiment analysis

- Automatically identifying opinions about a product, brand, or social issue

Applications

Sentiment analysis

- Automatically identifying opinions about a product, brand, or social issue



Applications

Sentiment analysis

- Automatically identifying opinions about a product, brand, or social issue



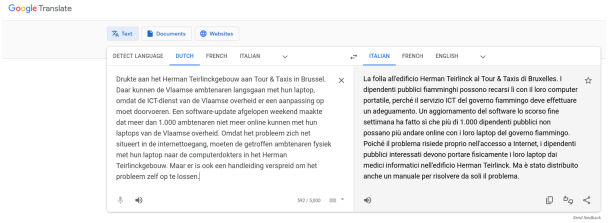
versus



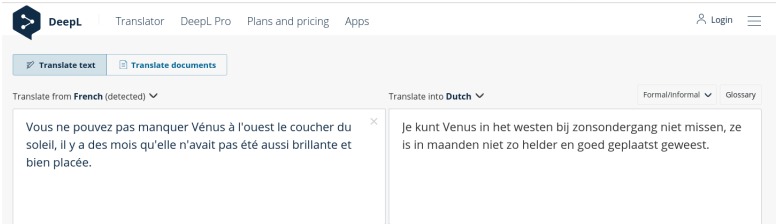
Applications

Translation

translate.google.com



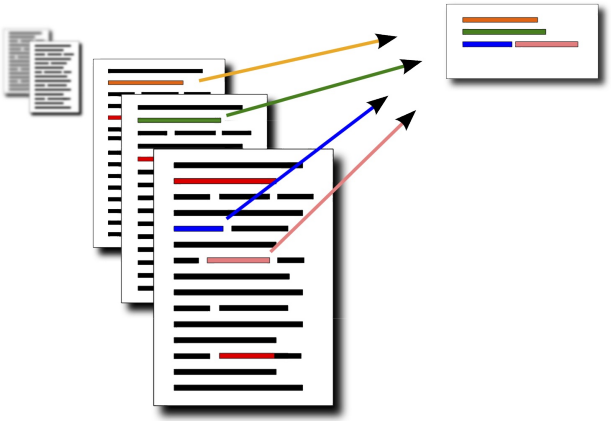
www.deepl.com



Applications

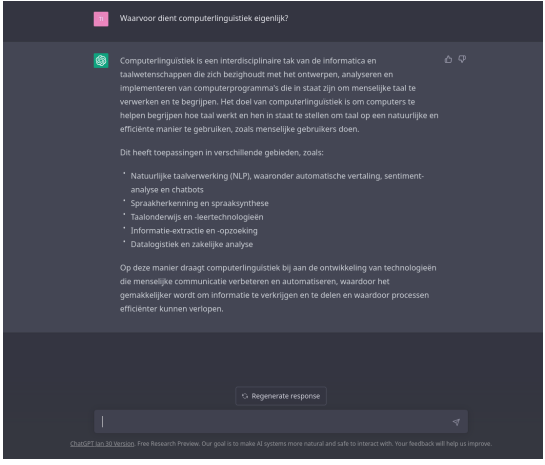
Summarization

- Synthesis of multiple documents
- Simplification



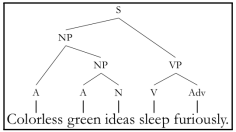
Applications

Large language models (ChatGPT)



The role of linguistics

- How can linguistic phenomena be modeled computationally within the three aforementioned paradigms?
  - how to model words?
  - how to model sentences?
  - construction of meaning representation
- How might linguistic theory be harnessed to improve natural language processing applications?





Learning Objectives

- Theoretical
  - Fundamental concepts of computational linguistic analysis / natural language processing
  - Overview of key research questions, approaches, and applications
- Practical
  - Basic programming skills for linguistic applications in Python
  - Applying existing algorithms and models using existing language processing modules
- Critical reflection
  - Limitations and societal impact of language technology (bias, interpretability)

Python

- ‘hands on’
- Programming exercises using the Python programming language
- Prior programming knowledge is useful
- Self-study ‘Introduction to Python for Linguists’ available as an interactive notebook

Notebook

- Environment that allows combining text, program code, and results
- Code can be executed interactively
- Two options:
  - Installation on your own computer (Jupyter notebook)
  - Google Colab web interface (Google account required)

Python: installation

- Standard version of Python
  - Available at `http://www.python.org`
  - Can be somewhat cumbersome to use, especially on Windows
- Anaconda (recommended)
  - Available at `https://www.anaconda.com/products/individual`
  - User-friendly distribution
  - Suitable for *data science* and machine learning
  - Easy package management
- Note: the installation process (which installer to download) depends on your operating system

Practical Information

- Full course content (slides, additional reading) is available **online via Toledo**
- Lectures are recorded (if infrastructure allows) and made available online







Evaluation

- No exam
- Final project
- Applying acquired knowledge and skills to a self-chosen NLP application or (computational) linguistic research question
- Note: 4 ECTS and 6 ECTS version



Topics

- Introduction
- Machine learning for language
- Neural networks for NLP
- Language modeling
- Grammar models
- Semantics
- Attention and transformers
- Machine translation, multilinguality, multimodality
- Interpretability and bias
- The role of linguistics
- Applications in linguistics and the humanities (only 6 ECTS)

Language use: an example

– Why did the chicken cross the road?
– I give up.

– To get to the other side.



Language use: an example

– **Why did** the chicken cross the road?
– I give up.

– To get to the other side.



Language use: an example

– **Wide id** the chicken cross the road?
– I give up.

– To get to the other side.



Language use: an example

– Why did the **chicken cross** the road?
– I give up.

– To get to the other side.



Language use: an example

– Why did the chicken **cross** the road?
– I give up.

– To get to the other side.



Language use: an example

– Why did the chicken **crossed** the road?
– I give up.

– To get to the other side.



Language use: an example

– Why the chicken **did cross** the road?
– I give up.

– To get to the other side.



Language use: an example

– Why did the chicken cross the road?
– **I give up.**

– To get to the other side.



Language use: an example

– Why did the chicken cross the road?
– I give up.

– **To get to the other side**.



What is language?

- Language is a communication system
- What is the key element in a communication system?



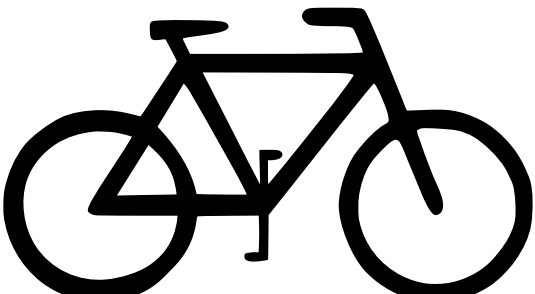


What is language?

- Language is a communication system
- What is the key element in a communication system?
- A **sign**
  - pairing of a form and a meaning
  - can be iconic, indexical, or symbolic











apple

penguin

hypothesis

What is language?

- Language is a communication system
- What is the key element in a communication system?
- A **sign**
  - pairing of a form and a meaning
  - can be iconic, indexical, or symbolic





What is language?

- Language is a communication system
- What is the key element in a communication system?
- A **sign**
  - pairing of a form and a meaning
  - can be iconic, indexical, or symbolic
- Language is a means of communication based on a system of signs





Animal vs. human communication

- Features of human language
  - Interchangeability
  - Cultural transmission
  - Arbitrariness
  - Discreteness
  - Displacement
  - Productivity
- Some are shared with animal communication systems, while others are (as far as we know) exclusively human



Vervet monkey

- Uses arbitrary signs in order to signal predators
- Not just utterances of stress or excitement
- Signs have stable external reference and serve the purpose of communication
- Three signs for different predators:
  - short, two syllable coughing sound
  - barking/peeping sound
  - continuous clicking, ‘chutter’ sound



Vervet monkey

- Uses arbitrary signs in order to signal predators
- Not just utterances of stress or excitement
- Signs have stable external reference and serve the purpose of communication
- Three signs for different predators:
  - short, two syllable coughing sound
    presence of an eagle → looking up and seeking shelter
  - barking/peeping sound
    presence of a leopard → escape in the trees
  - continuous clicking, 'chutter' sound
    presence of a snake → stand up and look down

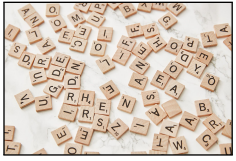


Human language: discreteness

- Signs are composed of discrete, meaningless units, which can be combined in various ways to create new signs with different meanings

  `stop` [s·t·o·p] → `spot` `tops` `opts`

- Animal communication systems mostly make use of signs that are indivisible units
- This feature of human language is also called **duality of patterning** or **double articulation**



Human language: displacement

- Humans can talk about things not present in space or time

  last night, *a man on the moon*, *a unicorn*, *in September 2037*, ...

- Animal communication systems are restricted to the present, the here and now



Human language: productivity

- Humans can produce (and understand) an infinite amount of utterances
  apple, *the apple*, *the green apple*, *the green apple that I ate last night*, *the green apple that I ate last night when I was...*
- in novel and creative ways
  e-mail, *brexit*, *tiktokification*, ...
- PHONEMES → MORPHEMES → WORDS → PHRASES → SENTENCES
- Animal communication: one utterance, one sign (vervet monkey)

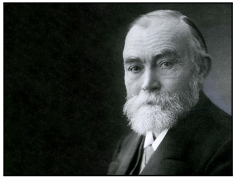


Compositionality

- Principle of compositionality (Frege): the meaning of an utterance is derived from the meaning of the signs and the way those signs are combined

  (1) Jim drives the car.

  (2) The car drives Jim.



- Co-compositionality: the influence that signs exert on each other when they are combined

  (3) red rose

  (4) red wine

  (5) red hair

How to make sense of human speech?

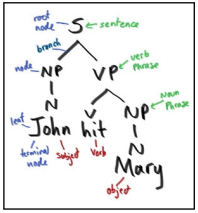
- Segmenting the utterance in terms of the basic units of sound and meaning
- Recognizing how the basic units are composed into larger units (syntax)
- Deriving the meanings of complex signs (compositional semantics)
- Contextualizing the meanings so that full interpretations arise

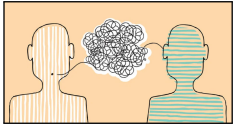




Linguistics

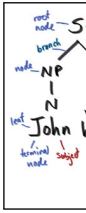
- Linguistics is the study of language
- Linguistics is concerned with all aspects of how humans use language, and what they must know in order to do so
- Linguistics asks different types of questions:
  - How are languages structured?
  - How are languages used?
  - How do languages change?

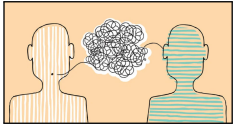




Linguistics

- Linguistics is the study of language
- Linguistics is concerned with all aspects of how humans use language, and what they must know in order to do so
- Linguistics asks different types of questions:
  - **How are languages structured?**
  - How are languages used?
  - How do languages change?





A brief history of linguistics

- India: description of Sanskrit (Panini)
- Greece/Rome: description of ancient Greek (Latin) for rhetorical purposes
- Middle ages: 7 arts – *trivium* (grammar, logic, and rhetorics) & *quadrivium* (arithmetic, geometry, astronomy and music)
- Renaissance: attention for other languages than Greek and Latin; standardization of speech and writing

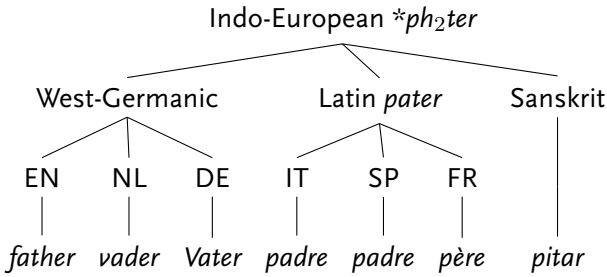


Birth of linguistics as a science

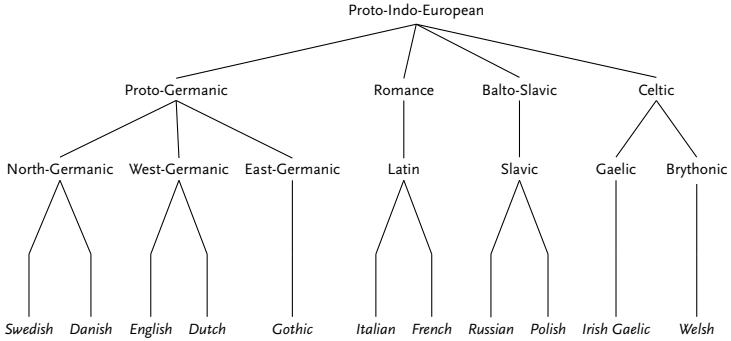
- 1786: William Jones hints at common root of Sanskrit, Greek, and Latin, and other European languages
- 19th century : historical and comparative linguistics (Brothers Grimm)







The Indo-European Language Family



First half of the 20th century

- Structuralism
  - Ferdinand de Saussure
  - *signifiant* vs. *signifié*, arbitrariness of the sign
  - the meaning of signs is determined by the other signs within the language structure
- Descriptive linguistics
  - Edward Sapir, Benjamin Whorf, Leonard Bloomfield
  - study of native American languages, with emphasis on the differences
  - linguistic relativity: grammatical systems and language use affect perception

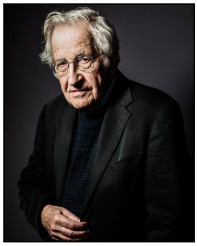






Noam Chomsky

- Generative grammar
  - Grammar as a system of rules that generates the grammatical sentences (and only those) for a given language
  - competence vs. performance
  - linguistic universals
  - need for formalization and full explicitness
- Language acquisition device
  - discrepancy between
    - apparent complexity of natural languages
    - ease with which children acquire them
  - language is innate



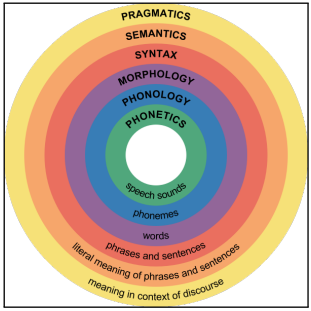
Relevance for our course

- Computational linguistics has been greatly influenced by Chomskyan formalization
- More recently: implicit revisiting of earlier theories (structuralism)
- What is it that makes language difficult for computers?
- A complex phenomenon that involves the interaction of various types of knowledge and skills
- Language understanding has to cope with an ubiquitous property of natural languages: **ambiguity**



Levels of linguistic structure

- phonology: sounds
- morphology: structure of words
- syntax: structure of sentences
- semantics: the meaning of words, and how they combine to form larger meanings
- pragmatics: the meaning of utterances within a discourse context



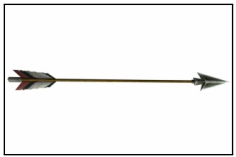
Ambiguity: phonology

- homophones : buy, by, bye
- important for speech recognition



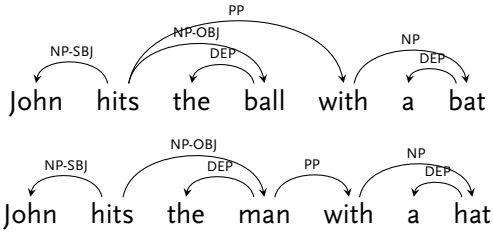
Ambiguity: morphology

- homonyms/homographs:
  - flies (noun) / flies (verb)
  - past (preposition) / past (noun)
  - like (preposition) / like (verb)
- complicates syntactic analysis
- complicates semantic analysis
- when combined, the ambiguity becomes even more important
  - Time flies like an arrow.
  - Fruit flies like a banana.



Ambiguity: syntax

Structural ambiguity







Ambiguity: semantics

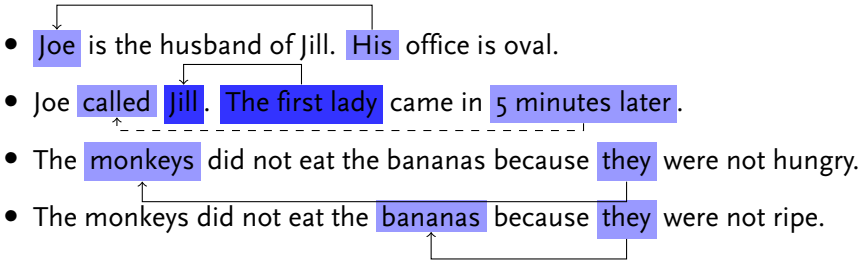
- Leslie used a *pen* to write this letter.
- We put he pigs in a *pen*.
- The judge sent them to the *pen* for a decade.
- They should *pen* that letter quickly.







Ambiguity: pragmatics



Difficulties

Machine translation

- word order in English is *modifier-noun*
  word order in French is *noun-modifier*
- English: *a red house*
  French: *une maison rouge*
- English: *almost double the usual oatmeal biscuits*
  French: *presque le double des biscuits à l'avoine habituels*





Difficulties

Machine translation

- Word can have different meanings
- e.g. French *avocat*

Difficulties

Machine translation

- Word can have different meanings
- e.g. French *avocat*



Difficulties

Machine translation

- Word can have different meanings
- e.g. French *avocat*





Difficulties

Machine translation

- Word can have different meanings
- e.g. French *avocat*



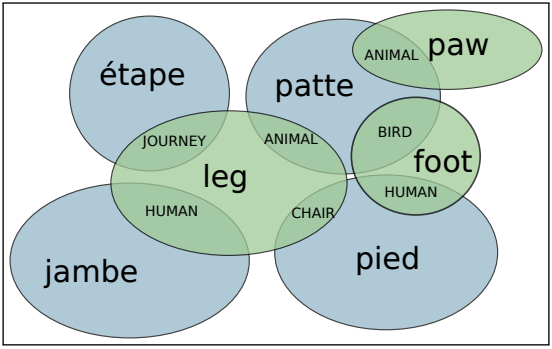
avocado



lawyer

Difficulties

Machine translation



Difficulties

Machine translation

- Certain languages, like Spanish, can omit subject pronouns
  - –o = I
  - –as = you
  - –a = he/she/it
  - –amos = we
  - –an = they
- When should the system use *he, she, it*?



Difficulties

Machine translation

- The monkeys did not eat the bananas because *they* were not hungry.
- Les singes n'ont pas mangé les bananes parce qu'*ils* n'avaient pas faim.
- The monkeys did not eat the bananas because *they* were not ripe.
- Les singes n'ont pas mangé les bananes parce qu'*elles* n'étaient pas mûres.



Difficulties

Machine translation

- I saw *her* yesterday.
- Ich habe *sie* gestern gesehen.
- I gave *her* a present.
- Ich habe *ihr* ein Geschenk gegeben.
- I also gave *her* brother a present.
- Ich habe auch *ihrem* Bruder ein Geschenk gegeben.



Difficulties

Machine translation

- French: *je bois du lait*



Difficulties

Machine translation

- French: *je bois du lait*
- English: *I drink milk* ?



Difficulties

Machine translation

- French: *je bois du lait*
- English: *I drink milk* ?
- English: *I'm drinking milk* ?



Difficulties

Machine translation

- *bone of contention*
- *kick the bucket*
- *Don't sell the bear's fur before hunting it.*







Difficulties

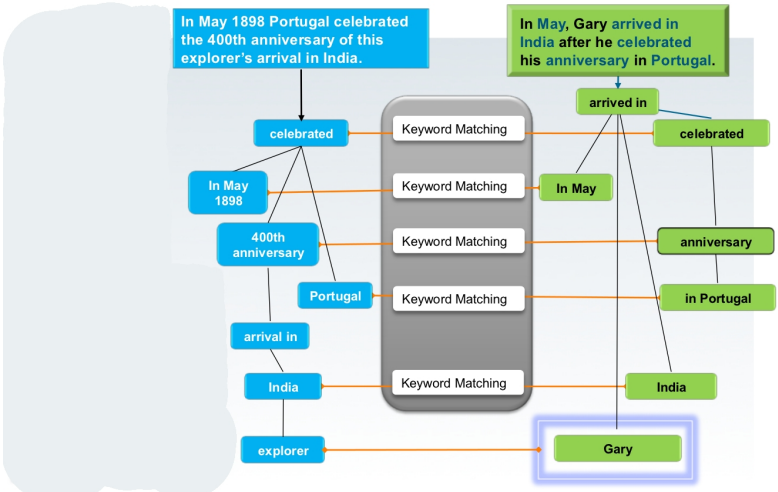
Spell-checking

- * This is too hot too drink.
- * This is to hot to drink.
- * This is to hot too drink.
- This is too hot to drink.



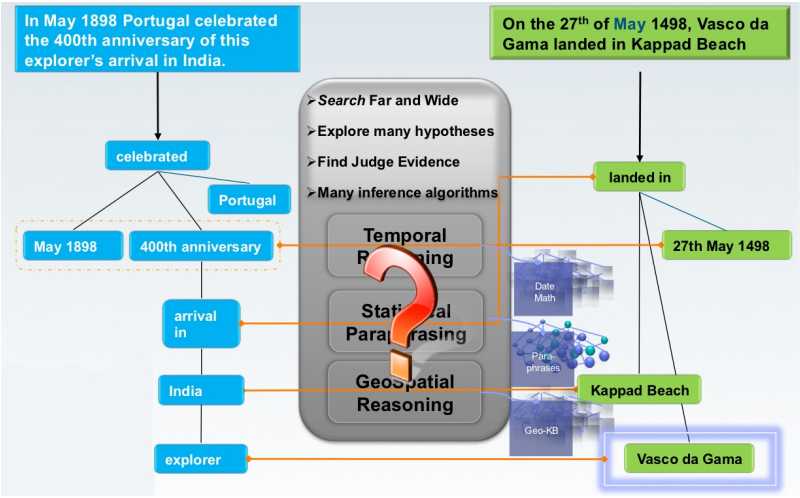
Difficulties

Question-Answering



Difficulties

Question-Answering



Difficulties

Speech technology

- Speech Recognition
  - fish vs. ghoti
  - The system is trained to wreck a nice peach
- Speech Synthesis
  - conténт (adjective) vs. cóntent (noun)
  - addréss (verb) vs. áddress (noun)







Difficulties

Speech technology

- Speech Recognition
  - fish vs. ghoti rough women nation
  - The system is trained to wreck a nice peach
- Speech Synthesis
  - conténts (adjective) vs. cóntent (noun)
  - addréss (verb) vs. áddress (noun)







Difficulties

Speech technology

- Speech Recognition
  - fish vs. ghoti rough women nation
  - The system is trained to recognize speech
- Speech Synthesis
  - conténts (adjective) vs. cóntent (noun)
  - addréss (verb) vs. áddress (noun)





